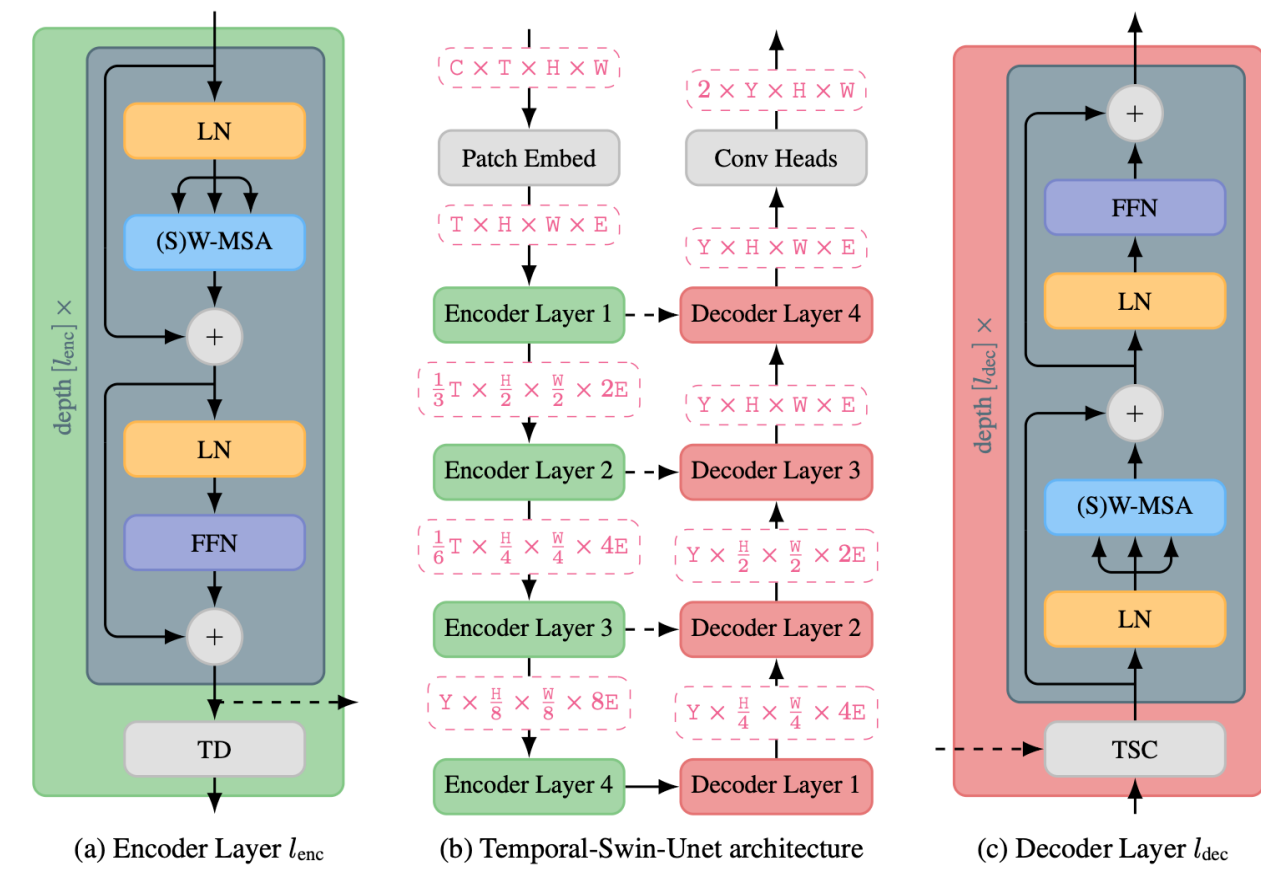


Tree Canopy Height

Jan Pauls, Karsten Schröder, Sven Lignsa, Martin Schwartz, Berkant Turan, Max Zimmer, Sassan Saatchi, Sebastian Pokutta, Philippe Ciais, Fabian Gieseke

- Above-ground biomass (AGB) is one of the key variables in climate policy decision making
- Tree height is used as proxy due to large availability of data
- Yet no temporal canopy height map exists at global scale

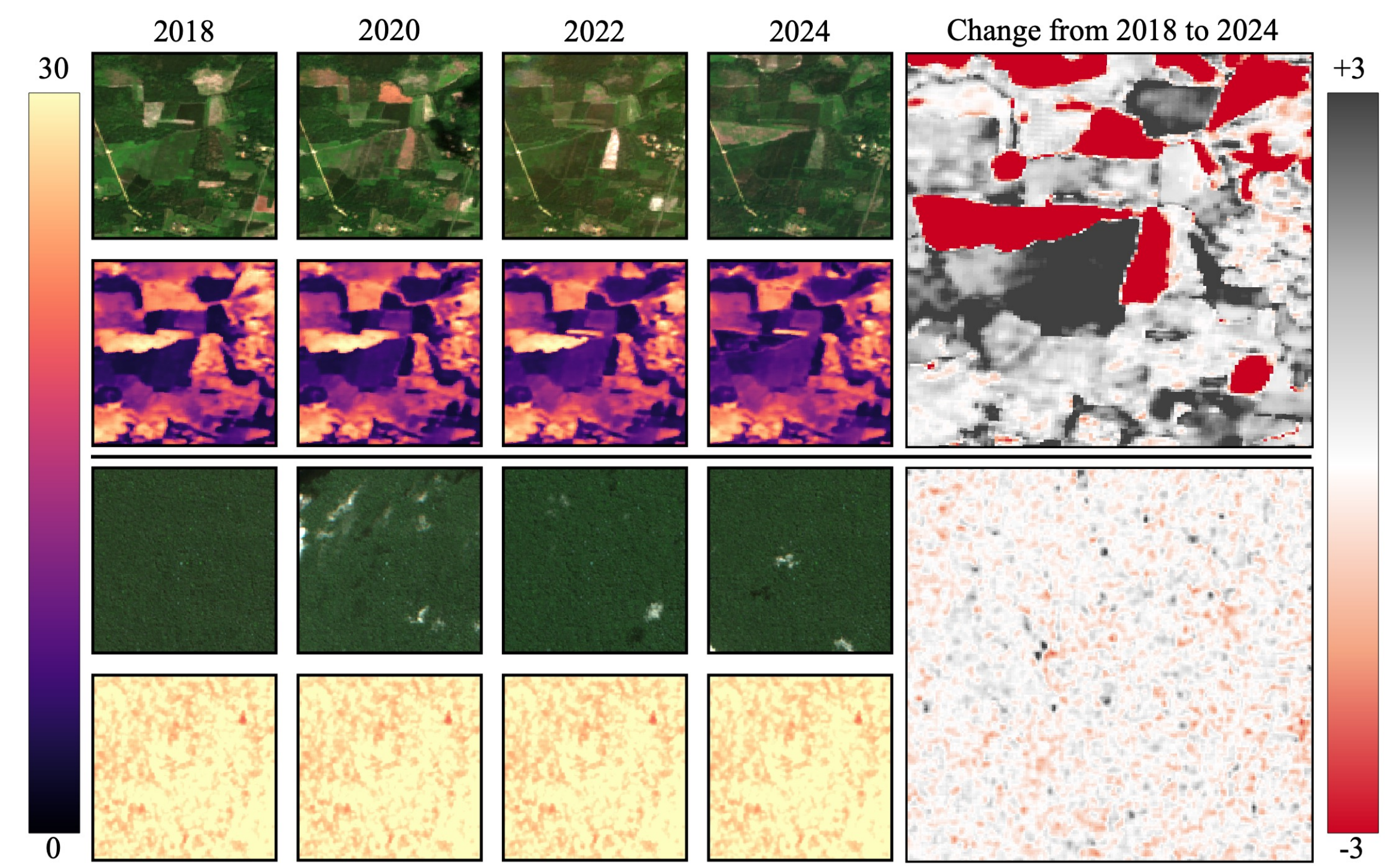
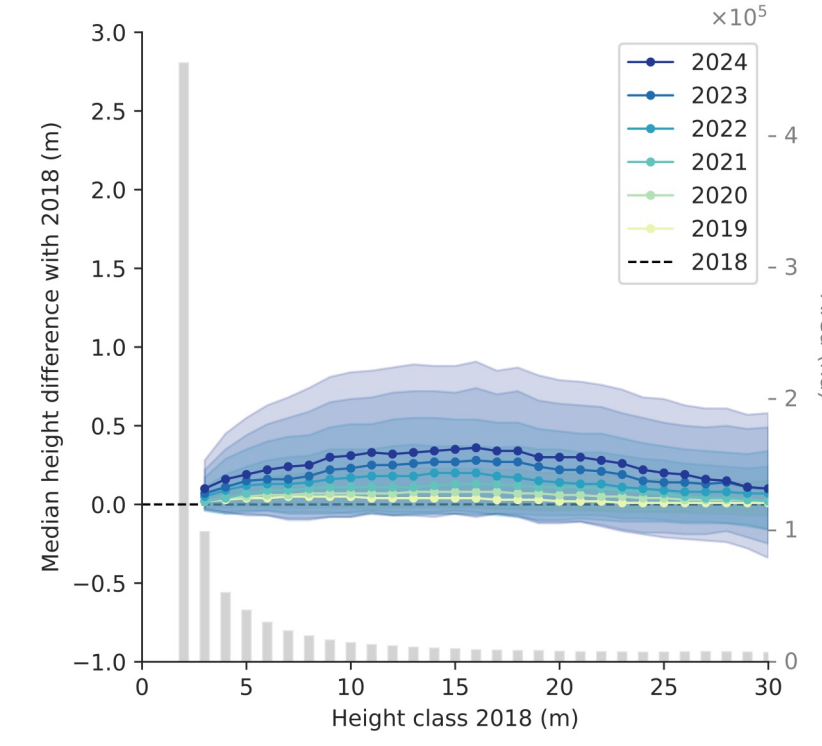


- we use a SwinVideoUnet
- modified TemporalDownsample (TD) and TemporalSkipConnection (TSC) to handle monthly input to yearly output conversion

Table 1: Comparison of global-scale methods for 2020 regarding MAE (m), MSE (m^2), RMSE (m), MAPE (%), R^2 and R^2_{all} (on all labels, including labels below 5 m). IQR is given in square brackets.

Method	MAE ↓	MSE ↓	RMSE ↓	MAPE ↓	R^2 ↑	R^2_{all} ↑
Potapov et al. (2021)	9.11 [7.73]	185.52 [107.99]	13.62 [7.73]	54.90 [78.99]	0.50	0.70
Lang et al. (2023)	7.97 [6.67]	143.78 [81.04]	11.99 [6.67]	53.33 [76.84]	0.52	0.71
Pauls et al. (2024)	6.85 [6.41]	138.19 [60.75]	11.76 [6.41]	34.20 [33.16]	0.51	0.73
Tolan et al. (2024)	11.89 [9.09]	260.28 [182.25]	16.13 [9.09]	68.78 [57.80]	0.45	0.64
Ours	5.85 [4.90]	118.07 [36.16]	10.87 [4.90]	30.20 [33.32]	0.59	0.77

- Input data:
- Monthly Sentinel-2
- Quarterly Sentinel-1
- Yearly ALOS Palsar-2
- One-time digital elevation model (DEM)
- One-time Forest/Non-Forest Map (FNF)



Website



AI4Forest

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AI4Forest is dedicated to revolutionizing the monitoring of forest ecosystems by fusing advanced Artificial Intelligence with high-resolution Earth observation. Our mission is to develop scalable, energy-efficient AI models capable of processing massive satellite datasets to generate precise, weekly or monthly updated maps of forest health, biomass, and carbon dynamics—down to the level of individual trees.

By overcoming the limitations of current monitoring systems, this French-German collaboration aims to provide decision-makers with the actionable, near real-time data needed to mitigate climate change impacts and support sustainable forest management, all while fostering open science and interdisciplinary excellence.



- Basile et al. (2025). Remote-sensing-based forest canopy height mapping: some models are useful, but might they provide us with even more insights when combined? *Geoscientific Model Development*, 18(2), 337–369. doi:10.5194/gmd-18-337-2025
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- Schwartz et al. (2024). High-resolution canopy height map in the Landes forest (France) based on GEDI, Sentinel-1, and Sentinel-2 data with a deep learning approach. *International Journal of Applied Earth Observation and Geoinformation*.

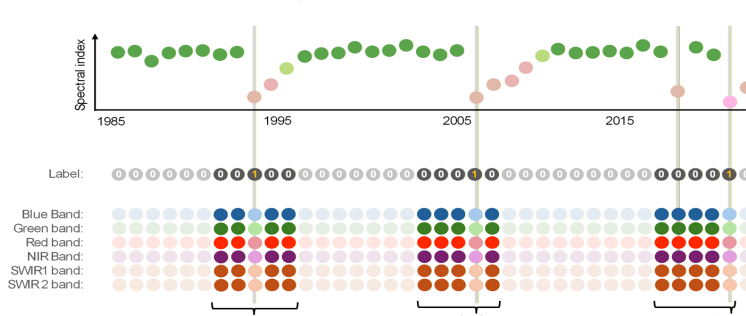


Disturbances

- Climate extremes and land-use change increasingly stress European forests
- Understanding forest disturbances requires long-term monitoring
- Continental-scale, consistent mapping remains a key challenge

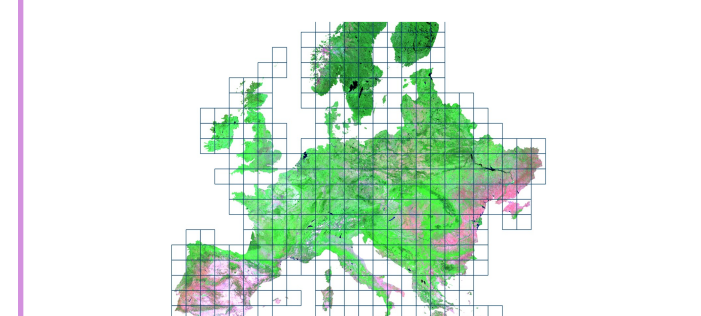
Background

- Forest disturbances are defined as loss of 50 % of canopy cover
- Multiple agents: fire, harvest, insects, storm...



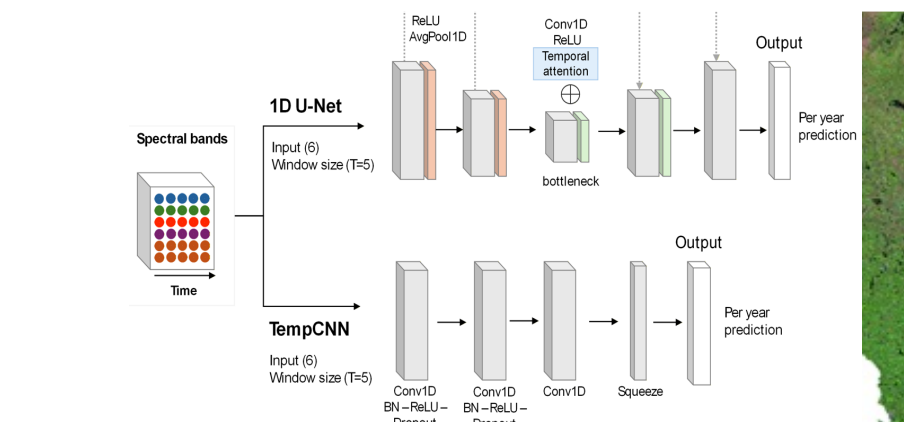
Data

- Landsat BAPs
- 20,084 randomly sampled points x 33 years in Europe
- Input: RGB, NIR, SW1, SW2
- 2 % disturbance labels

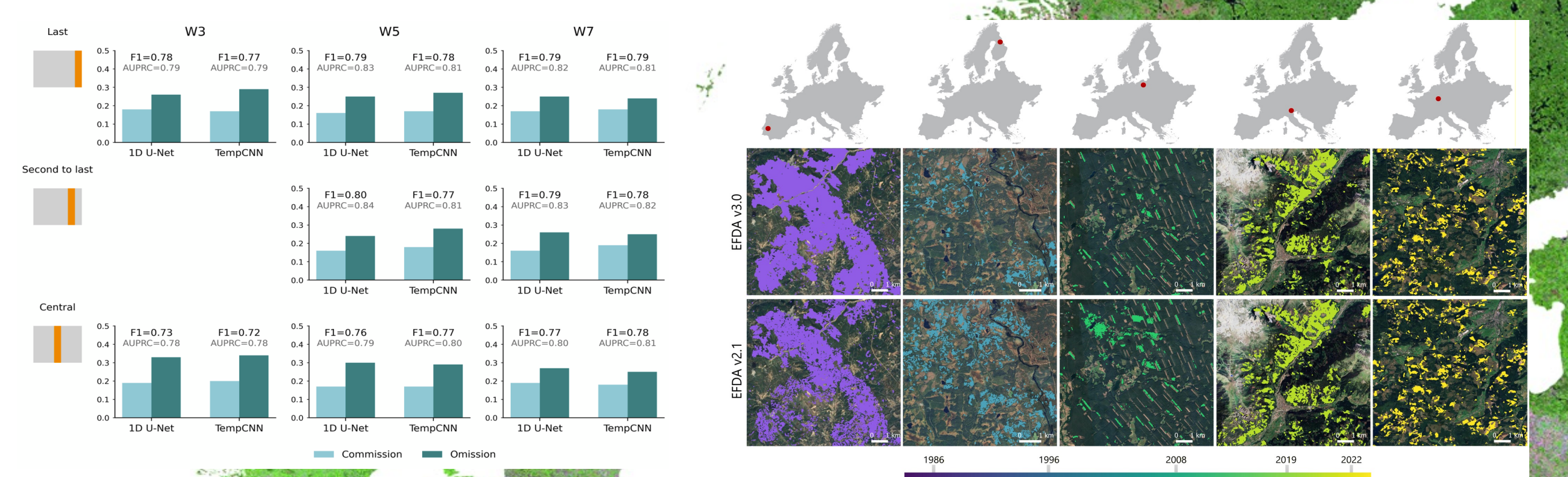


Model

- Multi-scale temporal kernels
- Focal loss for class imbalance
- Temporal attention + sliding window



Results



Conclusion

- Temporal convolutional architectures are well suited for disturbance detection
- Short wave infrared and near-infrared bands dominate model performance
- Balanced commission and omission errors
- Visually smoother maps



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